

If at least one of  $m$  or  $n$  is 0, then  $mn=0 \therefore mn \neq 1$ . So,  $\neg p$  is True.  
By defn. of negation,  $\neg(\neg p)$  is False. By Double-Negation Law,  $p$  is False.  
By Defn. of AND,  $(p \wedge q \wedge r)$  is False. By defn. of negation,  $\neg(p \wedge q \wedge r)$  is True.

$\therefore$  Let,  $m \neq 0$  and  $n \neq 0$ . Let,  $m$  not be an integer.  $\therefore \neg q$  is True. By defn. of negation,  $\neg(\neg q)$  is False. By Double-negation Law,  $q$  is False. By defn. of AND,  $(p \wedge q \wedge r)$  is False. By defn. of negation,  $\neg(p \wedge q \wedge r)$  is True.

Let,  $n$  not be an integer.  $\therefore \neg r$  is True. By defn. of negation,  $\neg(\neg r)$  is False. By Double-negation Law,  $r$  is False. By defn. of AND,  $(p \wedge q \wedge r)$  is False. By defn. of negation,  $\neg(p \wedge q \wedge r)$  is True.

$\therefore$  Let both  $m$  and  $n$  are integers.

Case i)  $n \geq 1$  and  $m \geq 1$ .  $\therefore mn \geq 1 \cdot 1 = 1 \therefore mn \neq 1$

Case ii)  $n \geq 1$  and  $m = -1$ .  $\therefore mn \leq -1 < 1 \therefore mn \neq 1$

Case iii)  $n < 0$  and  $m \geq 1$ .  $\therefore mn < 0 < 1 \therefore mn \neq 1$

Case iv)  $n < 0$  and  $m = -1$ .  $\therefore mn > 1 \therefore mn \neq 1$

In each of the 4 cases,  $mn \neq 1$ .  $\therefore \neg p$  is True. By defn. of negation,  $\neg(\neg p)$  is False. By Double-negation Law,  $p$  is False. By the defn. of AND,  $(p \wedge q \wedge r)$  is False. By the defn. of negation,  $\neg(p \wedge q \wedge r)$  is True.

**Case 3:** Let,  $(\neg s \wedge \neg v)$  be True. By defn. of AND,  $\neg s$  is True and  $\neg v$  is True.  
 $\therefore m \neq 1$  and  $n \neq -1$ . The argument is similar to Case 2.

**Case 4:** Let,  $(\neg t \wedge \neg v)$  be True. By defn. of AND,  $\neg t$  is True and  $\neg v$  is True.  
 $\therefore n \neq 1$  and  $n \neq -1$ . The argument is similar to Case 1

By defn. of conditional,  $\neg((s \wedge t) \vee (u \wedge v)) \rightarrow \neg(p \wedge q \wedge r)$  is True.  
 $\therefore (p \wedge q \wedge r) \rightarrow (s \wedge t) \vee (u \wedge v)$  is True.